TCP-UDP COMPARISON PROJECT

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Abstract

In this study, the performance of TCP and UDP protocols is analyzed in a simulated network environment operating under different Signal-to-Noise Ratio (SNR) values. For this purpose, a client–server based file transfer system was developed using the C programming language. Through a series of experiments, key network performance metrics such as goodput (Mbps), packet loss , packet corruption rate , and transfer time (s) were measured and compared.

During the simulation, the theoretical Bit Error Rate (BER) was calculated for various SNR levels, and the obtained results were compared with Shannon's Channel Capacity. While the TCP protocol ensures reliability by using built-in packet sequencing and retransmission (recovery) mechanisms, the UDP protocol was subjected to probabilistic packet losses in order to represent unreliable transmission conditions observed in real-world networks.

The results obtained at different SNR levels demonstrate the impact of channel quality on both connection-oriented (TCP) and connectionless (UDP) protocols, and clearly show how these protocols behave under varying network conditions.

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1 Introduction

In this project, the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP) are compared under various SNR levels. A client–server based file transfer application is implemented, and channel noise is simulated by introducing bit errors and packet losses according to theoretical models. For each SNR value, performance metrics such as transfer time, packet loss, corrupted packets and goodput are measured. The main goal of this study is to observe how TCP and UDP differ in terms of reliability and efficiency when channel quality changes. While TCP aims to provide reliable data delivery through retransmissions, UDP offers faster transmission at the cost of possible data loss. By analyzing the experimental results, the advantages and limitations of both protocols under noisy channel conditions are evaluated.

2 Procedure

The experimental procedure followed in this study is summarized as follows:

1. A client–server based file transfer application was developed using both TCP and UDP protocols. The server listens for incoming requests and transmits the requested file to the client, while the client receives the data and records performance statistics.

2. Channel noise was simulated by introducing bit errors and packet losses depending on the specified Signal-to-Noise Ratio (SNR). For TCP, corrupted packets were detected using checksum validation and retransmitted when necessary. For UDP, packet drops and limited retransmission attempts were implemented to simulate unreliable channel behavior.

3. Experiments were performed for a user-defined range of SNR values (start, end, and step). For each SNR value, the file transfer was repeated a fixed number of times (RUNS = 3) to obtain more stable average results. Although the number of repetitions was kept low to ensure faster program execution, the average of three runs was used to improve data robustness. The system also supports simultaneous testing with up to five clients.

4. During each experiment, key performance metrics such as total transfer time, number of lost packets, number of corrupted packets, and achieved goodput were measured separately for TCP and UDP.

5. In addition, the theoretical Bit Error Rate (BER) and Shannon channel capacity were calculated for each SNR value and compared with the experimental results.

3 Theory

Data transmission reliability and efficiency depend on the channel signal-to-noise ratio (SNR). In this study, channel noise effects are analyzed using theoretical BER and Shannon capacity, and TCP and UDP file transfer performance is experimentally evaluated under identical conditions.

3.1 Signal-to-Noise Ratio (SNR)

The Signal-to-Noise Ratio (SNR) is defined as the ratio of signal power to noise power and is usually expressed in decibels (dB). It is given by:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(\frac{P_s}{P_n} \right),$$

where P_s represents the signal power and P_n represents the noise power. As SNR increases, the impact of noise decreases, resulting in more reliable data transmission.

3.2 Bit Error Rate (BER)

Bit Error Rate (BER) represents the probability that a transmitted bit is incorrectly received due to channel noise. In this project, Binary Phase Shift Keying (BPSK) modulation over an Additive White Gaussian Noise (AWGN) channel is assumed. The theoretical BER for BPSK is expressed as:

$$\mathrm{BER} = Q\left(\sqrt{2 \cdot \mathrm{SNR}}\right),$$

where $Q(\cdot)$ denotes the Q-function. This theoretical BER is used to simulate bit-level errors in the transmitted data and to evaluate how noise affects packet integrity.

3.3 Shannon Channel Capacity

Shannon channel capacity defines the maximum achievable data rate for a given bandwidth and SNR, assuming error-free transmission. It is given by:

$$C = B \log_2(1 + \mathrm{SNR}),$$

where B represents the channel bandwidth. In this study, Shannon capacity is used as a theoretical upper bound and is compared with the experimentally measured goodput values obtained from TCP and UDP transmissions.

3.4 TCP and UDP Behavior in Noisy Channels

TCP and UDP differ significantly in the way they handle transmission errors. TCP ensures reliable data delivery using acknowledgments and retransmissions, which preserves data integrity at the cost of increased latency in noisy channels. UDP, on the other hand, does not guarantee reliable transmission and may experience packet losses under low SNR conditions. By combining theoretical concepts such as BER and Shannon capacity with experimental measurements, the performance differences between TCP and UDP under noisy channel conditions are clearly demonstrated.

4 Experimental Setup

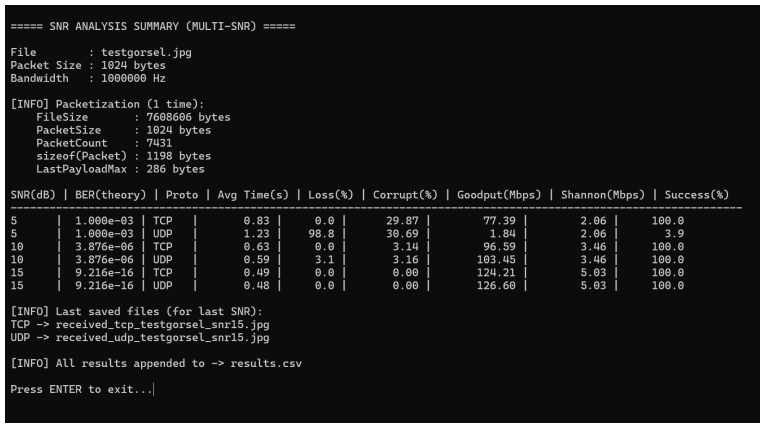
The experiments were performed using a client–server based file transfer system over TCP and UDP with a fixed packet size of 1024 bytes. Channel conditions were controlled by varying the Signal-to-Noise Ratio (SNR), and noise was modeled using a simplified Additive White Gaussian Noise (AWGN) channel based on theoretical Bit Error Rate (BER). For each SNR value, experiments were repeated three times (RUNS = 3) and averaged, with support for up to five simultaneous clients. The transmitted file was segmented into fixed-size packets, with the final packet possibly containing a smaller payload, enabling a fair comparison between TCP and UDP.

4.1 Protocol Operation Across Network Layers

- **Application Layer:** Responsible for file reading, segmentation into packets, and logging of transmission results. The client and server applications control the file transfer process.
- **Transport Layer:** TCP and UDP operate at this layer. TCP provides reliable transmission through sequencing, acknowledgments, and retransmissions, whereas UDP offers a connectionless service without reliability mechanisms.
- **Network Layer (Simulated):** Routing is abstracted in the simulation. Channel conditions such as packet loss and errors are modeled using an SNR-based approach.
- **Physical Layer (Modeled):** Noise effects are represented using an Additive White Gaussian Noise (AWGN) model, and bit-level errors are introduced based on theoretical Bit Error Rate (BER).

5 Results

This section presents the experimental results obtained from TCP and UDP file transfer tests conducted under different Signal-to-Noise Ratio (SNR) conditions. The results are summarized based on transfer time, packet loss, packet corruption, achieved goodput, and success rate.



```
===== SNR ANALYSIS SUMMARY (MULTI-SNR) =====

File        : testgorsel.jpg
Packet Size : 1024 bytes
Bandwidth   : 1000000 Hz

[INFO] Packetization (1 time):
    FileSize       : 7608606 bytes
    PacketSize     : 1024 bytes
    PacketCount    : 7431
    sizeof(Packet) : 1198 bytes
    LastPayloadMax : 286 bytes

SNR(dB) | BER(theory) | Proto | Avg Time(s) | Loss(%) | Corrupt(%) | Goodput(Mbps) | Shannon(Mbps) | Success(%)
-----------------------------------------------------------------------------------------------------------------
5       |  1.000e-03 | TCP   |       0.83 |     0.0 |      29.87 |        77.39 |        2.06 |     100.0
5       |  1.000e-03 | UDP   |       1.23 |    98.8 |      30.69 |         1.84 |        2.06 |       3.9
10      |  3.876e-06 | TCP   |       0.63 |     0.0 |       3.14 |        96.59 |        3.46 |     100.0
10      |  3.876e-06 | UDP   |       0.59 |     3.1 |       3.16 |       103.45 |        3.46 |     100.0
15      |  9.216e-16 | TCP   |       0.49 |     0.0 |       0.00 |       124.21 |        5.03 |     100.0
15      |  9.216e-16 | UDP   |       0.48 |     0.0 |       0.00 |       126.60 |        5.03 |     100.0

[INFO] Last saved files (for last SNR):
TCP -> received_tcp_testgorsel_snr15.jpg
UDP -> received_udp_testgorsel_snr15.jpg

[INFO] All results appended to -> results.csv

Press ENTER to exit...
```

Figure 1: Sample experiment results

At an SNR of 5 dB, the channel is highly noisy, resulting in a high theoretical BER. Under these conditions, TCP achieves a 100% success rate through retransmissions, at the cost of increased transfer time, while UDP suffers from severe packet loss and a very low success rate of 3.9%, highlighting its unreliability in noisy channels. When the SNR is increased to 10 dB, BER decreases significantly and both protocols show improved performance. TCP maintains reliable transmission with no packet loss, whereas UDP reaches a 100% success rate with minimal packet loss and slightly higher goodput. At a high SNR of 15 dB, channel conditions become nearly ideal and BER is negligible. Both TCP and UDP experience zero packet loss and corruption; however, due to lower protocol overhead, UDP achieves the highest goodput and the lowest transfer time. Overall, TCP ensures reliability under poor channel conditions, while UDP provides higher efficiency only when channel quality is high.

References

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